

BUSINESS ANALYTICS

INDIVIDUAL CASE STUDY

The Renewal Threshold: Predicting PG Resident Renewal Intention Using Behavioral and Service-Experience Analytics

Course: 23CSE452 – Business Analytics

Analytical Dataset: 10,000-row synthetic survey dataset

1. Problem Statement and Objectives

1.1 Problem Statement

The Paying Guest (PG) accommodation and co-living sector operates in a highly competitive environment where resident retention is an important contributor to long-term business sustainability. When residents leave after a relatively short stay, operators may incur repeated acquisition, onboarding, administrative, and vacancy-related costs. Consequently, understanding the factors associated with a resident's intention to renew is important for improving service quality and reducing avoidable churn.

PG operators traditionally focus on physical characteristics such as room quality, internet connectivity, air conditioning, furnishings, security, and other amenities to attract and retain residents. However, the resident experience extends beyond physical infrastructure. Operational factors such as the number of complaints raised, speed of complaint resolution, deposit refund transparency, and perceived fairness of rent can influence how residents evaluate the overall value of their accommodation.

This case study therefore investigates whether **service-experience variables provide greater predictive information about renewal intention than physical amenity and security variables**. The analysis compares three feature groups: an Amenity-only model, a Service-only model, and a Combined model.

The primary outcome is **Renewal Intention**, consisting of three survey responses: Yes, Maybe, and No. For binary predictive modeling, only definite Yes and No responses are considered, while Maybe responses are retained for descriptive analysis.

1.2 Business Significance

Resident retention has direct operational importance for PG management. If service-related experiences are strongly associated with renewal intention, management can use complaint and service information to identify residents who may require intervention.

For example, a complaint management system could identify residents who repeatedly report maintenance problems. Instead of waiting until the end of the rental period, management could intervene earlier by improving resolution speed, communicating progress, or escalating unresolved issues.

Similarly, understanding rent fairness and deposit transparency can help management improve the perceived value of the accommodation. Rather than focusing exclusively on physical upgrades, management can evaluate whether existing services are being delivered consistently and transparently.

Therefore, this study attempts to convert resident survey information into practical business intelligence that can support retention-oriented decision-making.

1.3 Objectives

The study has three major objectives:

- 1. **Descriptive and Exploratory Analysis:**
To examine the characteristics of the resident survey data and identify patterns between resident experience variables and renewal intention.
- 2. **Predictive Modeling and Feature Evaluation:**
To compare the predictive performance of Amenity-only, Service-only, and Combined feature blocks using Logistic Regression and Random Forest models.
- 3. **Business Recommendation:**
To translate analytical findings into practical recommendations related to complaint management, service quality, rent fairness, and resident retention.

1.4 Research Questions

The study addresses the following questions:

- Does service experience provide more predictive information about renewal intention than amenity and security ratings?
- Which resident experience variables are most strongly associated with renewal intention?
- Does combining service and amenity variables improve predictive performance?
- How can PG operators use these findings to design proactive resident-retention strategies?

2. Data Collection and Dataset Description

2.1 Data Collection

The analytical study uses a structured survey-style dataset containing 10,000 records. The dataset represents responses that could be collected through a questionnaire from PG residents.

The questionnaire structure captures demographic, financial, accommodation, service-experience, satisfaction, and renewal-intention information. The analytical file contains 12 principal fields representing different aspects of the resident experience.

The analysis is focused on understanding patterns in the supplied survey records rather than making claims about the entire PG accommodation population.

2.2 Dataset Characteristics

The dataset contains 10,000 records and 12 variables covering the following categories:

Category	Variable	Description
Demographic	Occupation	Student or working professional

Category	Variable	Description
Tenure	Stay Duration	Duration of residence in months
Financial	Monthly Rent	Current monthly rental amount
Physical Experience	Amenities Rating	Rating from 1–5
Physical Experience	Security Rating	Rating from 1–5
Service Experience	Complaint Count	Number of complaints/issues raised
Service Experience	Complaint Resolution Speed/status of complaint resolution	
Financial Experience	Deposit Refundability	Perception of deposit refund process
Financial Experience	Rent Fairness	Perception of whether rent is fair
Satisfaction	Overall Satisfaction	Overall resident satisfaction score
Target	Renewal Intention	Yes, Maybe, or No
Qualitative	Biggest Reason	Main reason stated by the resident

The variables allow the study to compare two broad dimensions of the resident experience:

Physical environment

- Amenities
- Security

Service and operational experience

- Complaint count
- Complaint resolution
- Deposit refundability
- Rent fairness

2.3 Renewal Intention Distribution

The primary target variable has three categories:

Renewal Intention Records Percentage

Yes	4,706	47.06%
Maybe	4,194	41.94%
No	1,100	11.00%
Total	10,000	100%

The largest individual category is **Yes**, representing 47.06% of the records. However, the large proportion of Maybe responses is also important because it indicates uncertainty among a substantial portion of residents.

For binary classification, Maybe responses are excluded. This produces a modeling dataset containing 5,806 definite Yes/No responses.

Within this subset:

- Yes = 4,706
- No = 1,100
- Yes proportion = approximately 81.05%
- No proportion = approximately 18.95%

This creates a class-imbalance problem. Consequently, class-balanced machine-learning models are used rather than relying on ordinary accuracy.

3. Data Preparation and Exploratory Analysis

3.1 Data Cleaning

The dataset was examined for missing values, duplicates, data types, and inconsistent categorical representations.

The analytical preparation included:

1. Checking missing values.
2. Checking duplicate records.
3. Converting numerical variables to appropriate numerical formats.
4. Standardizing categorical text values.
5. Encoding ordinal categorical variables.
6. Separating the three renewal categories for descriptive analysis.
7. Creating a binary Yes/No target for predictive modeling.
8. Excluding variables that could introduce target leakage.

The preprocessing was implemented using Python and common data-analysis and machine-learning libraries.

3.2 Encoding of Variables

Several categorical variables were converted into numerical representations for statistical analysis and machine learning.

Rent Fairness

The categories were represented in ordered form:

- Underpriced = 0
- Fair = 1
- Overpriced = 2

Deposit Refundability

The categories were ordered from lower to higher concern:

- Straightforward = 0
- Not sure = 1
- Disputes common = 2

Complaint Resolution

Complaint resolution was represented according to resolution speed:

- Same day = 0
- Few days = 1
- Over a week = 2
- Never resolved = 3
- No complaints raised = separate model category

This distinction is important because a resident with no complaint should not automatically be treated as a resident whose complaint was unresolved.

3.3 Target Leakage Prevention

Two variables require special treatment.

First, **Biggest Reason** is a free-text variable and was excluded from predictive modeling because it may directly contain the respondent's explanation for their renewal decision.

Second, **Overall Satisfaction** was excluded from the primary predictive feature comparison. Satisfaction can behave as a proxy for renewal intention and therefore may artificially inflate predictive performance.

Keeping these variables outside the primary feature blocks produces a more meaningful comparison of operational and physical experience variables.

3.4 Exploratory Analysis

Exploratory analysis was performed using frequency distributions, conditional renewal rates, descriptive statistics, and correlation analysis.

3.4.1 Amenity Rating

A positive relationship was observed between amenity ratings and renewal intention.

At an amenities rating of 2, approximately **32.4%** of residents expressed definite Yes renewal intention. At a rating of 5, this increased to approximately **54.9%**.

This indicates that physical amenities are relevant to renewal intention. However, the relationship is not sufficiently strong to suggest that amenities alone explain resident retention.

This observation motivates the comparison between physical features and service-experience variables.

3.4.2 Complaint Count

Complaint volume shows a stronger and more pronounced relationship with renewal intention.

The observed renewal rates were:

Complaint Count Definite Yes Renewal

0	64.3%
2	32.9%
4	10.6%

The pattern indicates a substantial reduction in definite renewal intention as complaint volume increases.

From a business perspective, this suggests that repeated service problems may act as an early warning signal. Complaint count can therefore be incorporated into a resident-retention monitoring system.

The relationship should nevertheless be interpreted as an association in the supplied analytical data rather than a causal effect.

3.4.3 Rent Fairness

Perceived rent fairness also shows a meaningful difference.

Residents who considered their rent **Fair** had a definite renewal rate of approximately **49.5%**, whereas residents who considered the rent **Overpriced** had a rate of approximately **20.2%**.

This indicates that residents evaluate accommodation not only according to the absolute rental amount but also according to their perception of the value received in exchange for that amount.

Consequently, rent communication, included services, deposit policies, maintenance, utilities, and other benefits can influence perceived value.

3.4.4 Security

Security ratings were also examined as part of the Amenity feature block. Security is an important component of residential accommodation, but the predictive comparison shows that the combination of service variables provides considerably more information than the physical Amenity-only block.

Security should therefore continue to be monitored as a basic operational requirement, while management should not assume that improvements in security or physical infrastructure alone will resolve retention problems.

3.5 Correlation Analysis

A Spearman correlation analysis was used for numerical and ordinal variables.

Spearman correlation was selected because several variables represent ordinal categories rather than continuous measurements.

The correlation analysis is used primarily for descriptive interpretation. The numerical codes assigned to categories represent ordering and should not be interpreted as equally spaced numerical quantities.

Correlation also does not establish causation.

4. Analytics Method and Implementation

4.1 Analytical Framework

The analytical process follows a structured Business Analytics workflow:

Data → Preparation → EDA → Statistical Analysis → Feature-Block Modeling → Model Evaluation → Feature Importance → Business Recommendations

The study combines descriptive analytics, inferential association testing, and predictive analytics.

4.2 Mann–Whitney U Test

The Mann–Whitney U test was used to compare numerical or ordinal feature distributions between definite Yes and No groups.

The test is suitable when variables are ordinal or when normality assumptions for parametric tests may not be appropriate.

Variables examined include:

- Amenities
- Security
- Complaint Count
- Complaint Resolution
- Deposit Concern
- Rent Unfairness
- Overall Satisfaction

A rank-biserial effect size was also calculated to provide an indication of the magnitude and direction of group differences.

4.3 Chi-Square Test

For categorical variables such as occupation, a Chi-square test of association was used.

The purpose was to determine whether the distribution of a categorical variable differed between residents expressing definite Yes and No renewal intention.

Cramér's V was also considered as an effect-size measure.

Because the dataset is synthetic, statistical significance should be interpreted cautiously. A large number of generated observations can produce very small p-values even when the practical importance of an association is limited.

4.4 Logistic Regression

Logistic Regression was used as an interpretable baseline predictive model.

The model estimates the probability of a binary outcome:

$$P(Y = 1) = \frac{1}{1 + e^{-z}}$$

where:

$$z = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_n X_n$$

In this study, $Y = 1$ represents definite Yes renewal intention.

Logistic Regression is useful because the direction and magnitude of coefficients can help explain how variables are associated with renewal probability.

4.5 Random Forest

Random Forest was selected as a non-linear ensemble model.

Unlike Logistic Regression, Random Forest can capture complex relationships and interactions between variables.

For example, the relationship between complaint count and renewal intention may differ depending on complaint resolution speed or perceived rent fairness. A tree-based ensemble can model such non-linear relationships.

Random Forest was therefore used as a complementary model to Logistic Regression.

4.6 Cross-Validation

The study uses **Repeated Stratified 5-Fold Cross-Validation with 10 repetitions**, producing 50 held-out validation folds.

Stratification ensures that the relative proportion of Yes and No observations is approximately maintained across folds.

The preprocessing operations are incorporated into the modeling pipeline so that transformations are performed appropriately within the cross-validation framework.

4.7 Class Imbalance

The definite Yes/No dataset contains approximately:

- 81.05% Yes
- 18.95% No

If a model predicted Yes for every resident, it would achieve approximately 81% accuracy despite completely failing to identify the No class.

Therefore, class-balanced learning strategies were used.

The study emphasizes:

- ROC-AUC
- Average Precision
- Balanced Accuracy
- F1-score

rather than relying exclusively on ordinary accuracy.

Balanced Accuracy

Balanced Accuracy can be represented as:

$$Balanced\ Accuracy = \frac{Sensitivity + Specificity}{2}$$

This provides a more informative evaluation when the classes are imbalanced.

5. Comparison with State-of-the-Art Methods

Recent studies related to tenant renewal, hospitality revisiting intention, and residential lease renewal were reviewed to position the current analysis within the broader research context.

Published Study / Year	Dataset	Method Used	Evaluation Metric	Key Result	Comparison with This Work
In (2024) – Commercial real-estate tenant renewal	Commercial real-estate tenant survey context	Influence/hypothesis testing	Statistical significance	Management reliability, adequacy and tenant empathy were associated with perceived value and renewal intention	Both studies examine service experience and renewal. The present study converts service dimensions into measurable predictive features and evaluates them using cross-valid